

ROLE-BASED ACCESS CONTROL (RBAC) MODEL

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1. Introduction to RBAC in BookExchange

Role-Based Access Control (RBAC) is a method of restricting system access to authorized users. For the future expansion of BookExchange, we have defined three primary roles to ensure data integrity and user safety on the University campus.

Role	Permissions / Access Levels
Guest (Unauthenticated)	• Browse all active book listings . • Search for books by title or course code. • View public "About" and "FAQ" pages.

Verified Student (User)	• All Guest permissions. • Create and post new book listings . • Manage personal listings (Edit, Delete, Mark as Sold) . • View seller contact information.
System Administrator	• All Student permissions. • Moderate/Delete any listing that violates campus policy. • Manage user account statuses (Suspend/Verify). • Access platform analytics and system logs.